ABDELRAHMAN SAMARA

Applied Data Science and AI Student

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SUMMARY

Third year Applied Data Science and AI student at Breda University of Applied Sciences, looking for an internship in software, machine learning or data. Experienced in taking machine learning models from prototype to production on client projects, including automated pipelines, testing and cloud deployment, and in leading a five person team as scrum master.

PROJECT EXPERIENCE

Client: NPEC | Team of 5 | April 2026 to June 2026

Plant Root Analysis Platform, Scrum Master

Production web app that locates root tips in petri dish scans, with automated retraining and deployment on Azure and on premise.

- Built the Azure ML training pipeline that registers a new model version only once it passes evaluation.
- Automated model selection with hyperparameter sweeps, tracking every run in MLflow and versioning the training data.
- Set up CI/CD with GitHub Actions: tests on every change, deployment to Azure and on premise, and canary rollouts.
- Turned my prototype into the core of the platform: a tested Python package for root segmentation and tip localisation.

Client: NPEC | Individual | November 2025 to January 2026

Root Detection and Robotic Inoculation System

- Built an end to end system that finds plant root tips in petri dish images and guides a simulated lab robot to inoculate them.
- Trained a deep learning segmentation model (F1 0.88) and a robot controller that hit all 15 root tips within 1 mm.
- Ranked 5th of 77 on the private Kaggle leaderboard for root length accuracy.

Client: Content Intelligence Agency | Team of 3 | September 2025 to October 2025

Emotion Detection Pipeline for TV Content

Pipeline that transcribes TV footage, translates it and labels each sentence with one of seven emotions.

- Selected the speech to text service by word error rate and trained an English to Dutch translation model (BLEU 34.7).
- Tested large language model prompting as an emotion classifier and built the data quality report.

Capstone case study, simulated fintech client | Team of 4 | May 2025 to June 2025

AI Stock Market Chatbot

Proof of concept chatbot that answers investor questions about NASDAQ-100 stocks.

- Trained the clustering model that groups the 100 companies by risk and the classifier that predicts price movement.
- Wrote the SQL scripts that build and populate the PostgreSQL database behind it.
- Restructured the codebase into a modular Python package managed with Poetry, together with a teammate.

OTHER INDIVIDUAL PROJECTS

- **Diabetes medication prediction:** regression, classification and clustering models, plus gradient descent built from scratch.
- **Responsible image classification app:** deep learning classifier with bias and explainability analysis and a user tested prototype.
- **Quantitative research study:** tested whether multi-class segmentation finds the root-shoot junction better than a binary model.
- **SDG progress dashboard:** interactive Power BI dashboard tracking a UN Sustainable Development Goal for the BUas SDG Hub.

SKILLS

Programming: Python, SQL, currently learning TypeScript and React

Machine learning: TensorFlow/Keras, scikit-learn, computer vision, NLP, large language models, reinforcement learning

Data: PostgreSQL, pandas, Power BI

Backend, cloud and DevOps: FastAPI, Django, Docker, Azure ML, MLflow, GitHub Actions, CI/CD

Ways of working: Git, Scrum, automated testing with pytest

EDUCATION

Bachelor in Applied Data Science and Artificial Intelligence

Breda University of Applied Sciences, Breda | 2024 to 2028 (expected)

ADDITIONAL INFORMATION

Languages: Arabic (fluent) | English (fluent) | Dutch (beginner, actively learning)

Availability: 18 January 2027 to June 2027